Mitochondrial Complex I Deficiency, Nuclear Type 39 (MC1DN39) — Disease Characterization Report

Autonomous literature/database-driven disease knowledge-base entry. Evidence source types are labeled: [Human clinical], [Model organism], [In vitro], [Computational/DB], [Review]. Because only 1–2 patients have been reported, most disease-level detail is necessarily extrapolated from the broader isolated nuclear complex I deficiency / Leigh syndrome literature and is explicitly flagged.

> **Important identification note.** MC1DN39 is caused by **NDUFB7** and corresponds to **OMIM #620135 / MONDO:0859320**. It must not be confused with **MC1DN32** (NDUFB8, OMIM #618252), which has adjacent nomenclature. This report was verified against EBI OLS4, ClinVar, MyGene.info, and Ensembl.

Summary (Answer to the Research Question)

Mitochondrial complex I deficiency, nuclear type 39 (**MC1DN39**; OMIM #620135) is an ultra-rare, autosomal recessive mitochondrial oxidative-phosphorylation disorder caused by **biallelic loss-of-function variants in *NDUFB7***, a nuclear-encoded accessory subunit of respiratory-chain **complex I** (NADH:ubiquinone oxidoreductase). Reduced NDUFB7 destabilizes complex I assembly/stability, producing **isolated complex I enzymatic deficiency**, impaired ATP synthesis, oxidative stress, and severe lactic acidosis. The single genetically and functionally proven case (P 33502047) presented with **intrauterine growth restriction, anemia, congenital lactic acidosis, hypertrophic cardiomyopathy, and encephalopathy, with a fatal neonatal/infantile outcome**; a homozygous intronic variant (c.113-10C>G) altered splicing, reduced NDUFB7

protein, and lowered complex I activity, and was rescued by wild-type NDUFB7 complementation. Prognosis is expected to be poor, consistent with the severe end of the nuclear complex I deficiency spectrum.

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1. Disease Information

- **What it is:** An inborn error of mitochondrial energy metabolism. Complex I (45 subunits in humans) is the entry enzyme of the respiratory chain; defects in its subunits/assembly factors are the most common cause of childhood mitochondrial disease [Review]. MC1DN39 is the subtype attributed to the accessory subunit *NDUFB7*.
- **Key identifiers (verified):**
- **OMIM (phenotype):** #620135 — "Mitochondrial complex I deficiency, nuclear type 39 (MC1DN39)"
- **OMIM (gene):** *NDUFB7* *603842
- **MONDO: MONDO:0859320**
- **MedGen / UMLS:** C5774258 (MedGen UID 1824031)
- **GARD:** 0026696
- **HGNC:** NDUFB7, HGNC:7702; **NCBI Gene (Entrez):** 4713; **Ensembl:** ENSG00000099795; **UniProt:** P17568
- **Gene locus: 19p13.12**; GRCh38 chr19:14,566,078–14,572,079, minus strand [Computational/DB — Ensembl REST + MyGene.info]
- **Orphanet:** subsumed under Isolated complex I deficiency (ORPHA:2609); no NDUFB7-specific ORPHA subtype
- **ICD-10:** E88.40 (mitochondrial metabolism disorder); **ICD-11:** 5C53.1 (disorders of mitochondrial energy metabolism)
- **MeSH:** "Mitochondrial Diseases" (D028361); "Leigh Disease" (D007888) for the encephalopathic spectrum
- **Synonyms / alternative names:** MC1DN39; complex I deficiency due to NDUFB7 deficiency; NDUFB7-related isolated complex I deficiency. NDUFB7 protein aliases: **NADH:ubiquinone oxidoreductase subunit B7, CI-B18, B18, NDUFB7 (B18 subunit)**.

- **Information source type:** Disease-level aggregated resources (OMIM/MONDO/ClinVar) plus a single detailed clinical case report (n=1 functionally proven; 1 additional ClinVar-listed variant). Individual/EHR-level data are minimal.

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2. Etiology

- **Primary cause (genetic):** Biallelic (homozygous or compound-heterozygous) pathogenic variants in *NDUFB7*. Causality established by whole-genome sequencing + RNA-seq splicing analysis + functional complementation [Human clinical/In vitro; P 33502047 : *"Complementation studies with expression of wild-type NDUFB7 in patient fibroblasts normalized complex I function."*].
- **Risk factors:**
- *Genetic:* The two biallelic *NDUFB7* alleles are the sufficient cause (Mendelian, high penetrance). Principal risk contexts are **carrier parents** and **consanguinity** (the reported patient carried a homozygous variant, consistent with autozygosity). No common-variant susceptibility loci (monogenic, ultra-rare; no GWAS).
- *Environmental:* None causal. Generic mitochondrial stressors (intercurrent infection/fever, fasting/catabolism, complex I–inhibiting drugs/toxins) can precipitate metabolic decompensation [Review — extrapolated].
- **Protective factors:** No genetic protective alleles known. Clinically, avoidance of catabolic stress and mitochondrial toxins is protective (extrapolated).
- **Gene–environment interactions:** Not specifically studied for *NDUFB7*; broadly, intercurrent illness/metabolic stress unmasks or aggravates the OXPHOS defect (extrapolated). Formal GxE epidemiology is not applicable to this monogenic disorder.

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3. Phenotypes

From the defining case (P 33502047) and the nuclear complex I deficiency spectrum (P 22644603):

Phenotype	Type	HPO term	Onset / severity / course	Frequency (reported)
Congenital/severe lactic acidosis	Lab abnormality	HP:0003128 (lactic acidosis) / HP:0002151 (elevated lactate)	Neonatal; severe; persistent	1/1
Hypertrophic cardiomyopathy	Sign/imaging	HP:0001639	Postpartum/ neonatal; severe	1/1
Intrauterine growth restriction	Sign	HP:0001511	Prenatal	1/1
Anemia	Lab abnormality	HP:0001903	Prenatal/ neonatal	1/1
Encephalopathy	Sign	HP:0001298	Neonatal; progressive	1/1
Neonatal onset / early death	Course	HP:0003623 (neonatal onset), HP:0003811 (neonatal death)	Neonatal/ infantile	1/1 (fatal outcome)
(Spectrum) hypotonia, developmental delay, brainstem/basal-ganglia lesions	Signs	HP:0001252, HP:0001263, HP:0002134	Infantile	Common across nuclear CI deficiency

- **Phenotype characteristics (overall):** Age of onset **prenatal–neonatal (congenital)**; severity **severe**; progression **progressive** with a **fatal** outcome in the reported case; frequency estimates are anecdotal (n=1) but consistent with severe complex I deficiency. Cardiac involvement (hypertrophic cardiomyopathy) is a prominent feature of this subtype [Human clinical; P 33502047].
- **Quality-of-life impact:** Profound — the reported presentation is a lethal neonatal multisystem disorder; survivors of comparable severe CI deficiency have severe

neurodevelopmental disability, feeding/respiratory dependence. No disease-specific EQ-5D/SF-36 data exist for this ultra-rare subtype.

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4. Genetic / Molecular Information

- **Causal gene:** ***NDUFB7*** (NADH:ubiquinone oxidoreductase subunit B7), HGNC:7702, OMIM *603842, 19p13.12; encodes a ~16–18 kDa accessory (supernumerary) subunit of complex I (UniProt P17568). NDUFB7/CI-B18 contains a CHCH (twin CX9C) domain** typical of intermembrane-space, disulfide-relay-imported proteins [Computational/DB — UniProt P17568].
- **Pathogenic variants (ClinVar, transcript NM_004146.6):**
- **c.113-10C>G** — intronic, creates a splicing defect; homozygous in the index patient; reduces NDUFB7 protein and complex I activity (functionally validated; the defining variant) [Human clinical; P 33502047].
- **c.311G>A (p.Arg104Gln)** — missense listed in ClinVar under "Mitochondrial complex I deficiency, nuclear type 39".
- ACMG/AMP classification: the c.113-10C>G variant is supported as **pathogenic/likely pathogenic** by rarity, functional RNA/protein/enzymatic data, complementation rescue, and genotype–phenotype fit; other listed variants require case-level curation.
- **Variant type/class:** Splice-altering intronic and missense alleles reported; loss-of-function is the operative mechanism.
- **Allele frequency:** Pathogenic alleles are extremely rare (absent/near-absent in **gnomAD**), consistent with a severe recessive disorder [Computational/DB — extrapolated]. **Gene-level constraint (gnomAD, ENSG00000099795):** NDUFB7 is **not constrained against loss of function** — pLI ≈ 0 (7.2e-8), oe_lof = 1.22 (90% CI 0.85–1.81; LOEUF ≈ 1.81), observed/expected LoF = 18/14.7, lof_z = −0.73. Heterozygous protein-truncating variants are tolerated, the expected signature of a **recessive, non-haploinsufficient** disease gene (carriers unaffected) [Computational/DB — gnomAD].
- **Origin: Germline**, biparental (recessive); not somatic.
- **Functional consequence: Loss of function** — reduced NDUFB7 → destabilized complex I assembly → isolated complex I deficiency [Human clinical/In vitro;

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P 33502047

mechanism corroborated by

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P 27626371

- **Modifier genes:** None identified for *NDUFB7*.
- **Epigenetic information:** None disease-specific (not available/applicable).
- **Chromosomal abnormalities:** None characteristic; single-gene disorder (not CNV/ aneuploidy). *Note:* Large chr19 duplications spanning NDUFB7 appear in ClinVar as unrelated cytogenetic events, not the cause of MC1DN39.

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5. Environmental Information

- **Environmental factors:** No causal environmental agent. Chemical complex I inhibitors (rotenone, MPTP/MPP+) model/aggravate CI dysfunction but do not cause this genetic disease [Review].
- **Lifestyle factors:** Not applicable to causation (congenital genetic disease); catabolic stress (fasting/illness) can trigger decompensation.
- **Infectious agents:** None; intercurrent infection acts only as a nonspecific metabolic stressor.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Biallelic *NDUFB7* variant (e.g., intronic c.113-10C>G) **leads to** aberrant *NDUFB7* mRNA splicing (RNA-seq confirmed). [Human clinical;

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   P 33502047
2. Aberrant splicing **results in** significantly **reduced NDUFB7 protein**. [Human clinical;

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   P 33502047

3. Loss of the NDUFB7 accessory subunit **leads to** destabilization of the membrane arm of complex I (accessory-subunit loss destabilizes its structural module). [In vitro; P 27626371]
4. This **results in** impaired **complex I assembly/stability → isolated complex I enzymatic deficiency**. [Human clinical; P 33502047]
5. Complex I deficiency **leads to** impaired NADH:ubiquinone electron transfer and reduced proton pumping → **decreased oxidative phosphorylation/ATP synthesis** and an elevated NADH/NAD+ ratio. [Review; P 34069703]
6. Redox imbalance/electron leak **results in** increased **mitochondrial ROS** and glutathione depletion (oxidative stress); severely CI-deficient cells may run F1Fo-ATPase in reverse to hold membrane potential, further depleting ATP. [In vitro; P 20157008 — inferred for NDUFB7 specifically]
7. ATP deficit **results in** compensatory anaerobic glycolysis → **severe lactic acidosis**. [Human clinical; P 33502047]
8. Energy failure preferentially injures **high-demand tissues → hypertrophic cardiomyopathy** (heart) and **encephalopathy** (CNS), with intrauterine growth restriction and anemia reflecting systemic bioenergetic failure. [Human clinical; P 33502047]
9. Progressive multisystem failure **leads to** early (neonatal/infantile) death in the reported case; (branch) milder alleles could, in principle, yield longer survival — untested. [Human clinical; P 33502047]

Inferred vs demonstrated: Steps 1–5, 7, 8 are demonstrated for NDUFB7; step 6 (ROS/reverse ATPase) is demonstrated in CI-deficient neuronal models and inferred here.

Supporting detail by category: - **Molecular pathways:** Oxidative phosphorylation / electron transport chain (KEGG hsa00190; Reactome "Complex I biogenesis" R-HSA-6799198). - **Cellular processes:** Bioenergetic failure, oxidative stress, cell death in high-demand tissues. GO: GO:0006120 (mitochondrial electron transport, NADH to ubiquinone); GO:0042773 (ATP synthesis coupled electron transport); GO:0034599 (cellular response to oxidative stress). - **Protein dysfunction:** Loss/reduction of a CHCH-domain accessory subunit → module destabilization (loss-of-function), not aggregation. Import likely via the mitochondrial **MIA40/CHCHD4 disulfide-relay** pathway (GO:0045041, protein import into mitochondrial intermembrane space) given the twin-CX9C motif [Computational/DB — inferred from UniProt P17568 domain]. - **Metabolic changes:** ↑ NADH/NAD+, ↑ lactate, ↓ ATP; energy-metabolism failure [Human clinical; P 33502047].

CHEBI: L-lactic acid (CHEBI:16651), NADH (CHEBI:16908), ubiquinone (CHEBI:16389). - **Immune involvement:** None primary. - **Tissue damage mechanisms:** Oxidative stress + energy depletion → cardiomyocyte hypertrophy/dysfunction and neuronal injury [Human clinical; P 33502047 ; Model organism P 26824698].

- **Biochemical abnormalities:** Enzymatic complex I (NADH:ubiquinone oxidoreductase) deficiency; reduced assembled complex I and reduced NDUFB7 on immunoassay [Human clinical; P 33502047].

- **Molecular profiling:** RNA-seq demonstrated the splicing defect (a key diagnostic modality here) [Human clinical; P 33502047].

- **GO/CL/UBERON suggestions:** GO:0032981 (mitochondrial respiratory chain complex I assembly), GO:1902600 (proton transmembrane transport), GO:0005747 (complex I), GO:0005743 (mitochondrial inner membrane), GO:0005758 (mitochondrial intermembrane space); CL:0000746 (cardiac muscle cell), CL:0000540 (neuron); UBERON:0000948 (heart), UBERON:0002420 (basal ganglia), UBERON:0001134 (skeletal muscle tissue).

7. Anatomical Structures Affected

- **Organ level — primary:** Heart (UBERON:0000948) — hypertrophic cardiomyopathy; brain (encephalopathy; basal ganglia UBERON:0002420, brainstem UBERON:0002298 in the broader spectrum). **Secondary/systemic:** hematologic (anemia), fetal growth (IUGR). **Body systems:** cardiovascular, nervous, hematologic, musculoskeletal.
- **Tissue/cell level:** Cardiomyocytes (CL:0000746), neurons/glia (CL:0000540); skeletal myofibers in the broader CI-deficiency spectrum. Tissue types: cardiac muscle, nervous tissue.
- **Subcellular level: Mitochondrion (GO:0005739)** — specifically the **inner membrane (GO:0005743)** (complex I) and **intermembrane space (GO:0005758)** (CHCH-domain of NDUFB7).
- **Localization / lateralization:** Cardiac involvement is global (biventricular hypertrophy); CNS lesions in the broader spectrum are characteristically **bilateral and symmetric** [Human clinical;

  P 33502047 ;

  P 22644603].

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8. Temporal Development

- **Onset:** Congenital — prenatal signs (IUGR, anemia) with neonatal decompensation (lactic acidosis, cardiomyopathy, encephalopathy); pattern acute-on-congenital [Human clinical;

  P 33502047].
- **Progression:** Rapidly progressive to a **fatal** outcome in the reported case; the broader nuclear CI-deficiency spectrum is progressive with high early mortality [PMID 22644603].
- **Patterns:** No remission in the reported case. Critical window for any intervention is very early. Duration: in the index patient the course was short/lethal; genotype-dependent variability is plausible but untested.

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9. Inheritance and Population

- **Epidemiology:** Ultra-rare — essentially a single functionally proven family (P 33502047) plus rare ClinVar entries; no prevalence/incidence estimates exist for MC1DN39 specifically. Context: Leigh syndrome incidence ≈ **1 in 77,000 live births** [Review; P 26363424]; isolated CI deficiency is the most common childhood mitochondrial biochemical defect.
- **Inheritance: Autosomal recessive** (biallelic *NDUFB7*) [Human clinical; P 33502047].
- **Penetrance:** Effectively complete for biallelic loss-of-function (high-penetrance recessive).
- **Expressivity:** Presumed variable across the CI-deficiency spectrum; too few cases to characterize for NDUFB7.
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects:** None reported for *NDUFB7*.
- **Consanguinity:** Homozygous variant in the index case is consistent with consanguinity/ autozygosity increasing recessive-disease risk.
- **Carrier frequency:** Not established; expected very low. gnomAD shows NDUFB7 tolerates heterozygous loss of function (pLI ≈ 0, LOEUF ≈ 1.81), consistent with asymptomatic carriers — as expected for a recessive gene [Computational/DB — gnomAD].
- **Population demographics:** No established ethnic predilection (single family). Expected sex ratio ~1:1 (autosomal 19p). Age distribution: neonatal/infantile.

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10. Diagnostics

- **Laboratory/biochemistry:** Marked **lactic acidosis** (elevated blood/CSF lactate); **spectrophotometric respiratory-chain enzyme assay** showing **isolated complex I deficiency**; anemia on hematology [Human clinical; P 33502047]. LOINC: lactate (2524-7).

- **Biomarkers:** Lactate; general mitochondrial-disease biomarkers GDF-15 and FGF-21 (extrapolated); **NDUFB7 protein abundance** by immunoassay is a specific readout.
- **Imaging: Echocardiography** for hypertrophic cardiomyopathy; **brain MRI** for encephalopathy/Leigh-like lesions; MR spectroscopy may show a lactate peak [Human clinical; P 33502047 ; spectrum P 22644603].
- **Muscle/fibroblast studies & histopathology:** Isolated CI deficiency biochemistry; BN-PAGE shows reduced assembled complex I; immunoblot shows reduced NDUFB7 [Human clinical; P 33502047].
- **Genetic testing (recommended approach): WGS or WES** is the primary route; NDUFB7 is intron-rich for splice variants, so **RNA-seq/transcriptomics is especially valuable** to detect and prove intronic splice-altering alleles (as in the index case) [Human clinical; P 33502047]. Mitochondrial/OXPHOS **gene panels** including *NDUFB7*; targeted single-gene confirmation with segregation. mtDNA testing excludes maternally inherited MT-ND causes. CMA/karyotype/FISH are not informative for this single-gene disorder.
- **Omics-based diagnostics:** RNA-seq (splicing); quantitative proteomics (loss of NDUFB7/module) — research-grade confirmation.
- **Clinical criteria & differential diagnosis:** Diagnosed within mitochondrial-disease criteria (isolated CI deficiency + biallelic *NDUFB7*). **Differential:** other nuclear CI-subunit/assembly-factor defects (incl. NDUFB8/MC1DN32, NDUFS/NDUFV subunits, NDUFAF assembly factors), mtDNA MT-ND mutations, other causes of neonatal lactic acidosis and infantile hypertrophic cardiomyopathy (e.g., other OXPHOS defects, Pompe disease, fatty-acid-oxidation defects). Distinguishing feature: isolated CI deficiency + *NDUFB7* genotype.
- **Screening:** Not in newborn screening. **Cascade carrier testing** of relatives and **prenatal/preimplantation genetic testing** are available once familial variants are known.

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11. Outcome / Prognosis

- **Survival/mortality:** Poor. The index NDUFB7 patient had a **fatal outcome** [Human clinical; 33502047]. Context (nuclear CI deficiency, n=130): **25% died before 6 months, >50% before age 2, 75% before age 10** [Human clinical; 22644603].
- **Morbidity/function:** Severe multisystem morbidity — cardiac failure, encephalopathy, refractory acidosis.
- **Disease course/complications:** Cardiac decompensation, recurrent/refractory lactic acidosis, respiratory failure, neurodevelopmental devastation.
- **Prognostic factors:** No reliable clinical/biochemical/genetic predictor of survival in the natural-history cohort [PMID 22644603]. Neonatal onset + cardiomyopathy generally portend the worst outcome (extrapolated).

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12. Treatment

No disease-modifying/curative therapy exists; management is supportive and symptomatic [Review; 28943110].

- **Pharmacotherapy (empiric "mitochondrial cocktail"):** riboflavin (vitamin B2; CHEBI:17015; NCIT C783), thiamine (CHEBI:18385; NCIT C934), coenzyme Q10/ubiquinol (CHEBI:46245; NCIT C1041), L-carnitine (NCIT C61735), biotin, antioxidants; management of lactic acidosis; heart-failure therapy for cardiomyopathy. Evidence base is weak (few controlled trials) [PMID 28943110].
- **Advanced therapeutics:** No approved gene/cell/RNA therapy for MC1DN39. Because wild-type *NDUFB7* complementation rescues patient fibroblasts in vitro [In vitro; 33502047], **gene replacement is a conceptual future avenue**; for the specific intronic splice variant,

splice-modulating antisense oligonucleotides are a theoretically attractive (experimental) strategy.

- **Surgical/interventional:** Supportive only (e.g., cardiac support, gastrostomy).
- **Supportive/rehabilitative:** Nutrition support, cardiology/neurology management, respiratory support, avoidance of mitochondrial-toxic drugs and catabolic stress; PT/OT/speech therapy in survivors.
- **Experimental:** Agents enhancing ETC function or mitigating consequences are studied for mitochondrial disease broadly (ClinicalTrials.gov); none NDUFB7-specific [Review; P 28943110].
- **Treatment outcomes:** Generally do not alter natural history [PMID 28943110].
- **Strategy / personalized medicine:** A trial of riboflavin/CoQ10 is reasonable; combination "cocktail" therapy is standard practice; genotype-guided splice-targeting is future-facing.

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13. Prevention

- **Primary prevention:** Not preventable (congenital genetic). **Genetic counseling** for at-risk/consanguineous families; carrier and cascade testing.
- **Secondary prevention:** Early diagnosis and metabolic-crisis prophylaxis (sick-day management; avoid fasting). Not part of population newborn screening.
- **Tertiary prevention:** Prevent complications — cardiac and respiratory surveillance, nutrition, infection prevention/immunization for intercurrent illness.
- **Reproductive options: Prenatal testing** and **preimplantation genetic testing (PGT)** for couples with known biallelic variants (ACMG/ACOG/NSGC frameworks).
- **Immunization / public health / environmental:** Standard childhood vaccination to reduce infection-triggered decompensation; no specific public-health or environmental intervention applies.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *NDUFB7* is conserved across metazoans (NCBI Taxon: *Homo sapiens* 9606; *Mus musculus* 10090 *Ndufb7*, NCBI Gene 66916, MGI:1914166; *Danio rerio* 7955; *Drosophila melanogaster* 7227 has complex I accessory-subunit homologs). Complex I accessory subunits are broadly conserved in eukaryotes.
- **Natural disease in animals:** No well-characterized spontaneous *NDUFB7* disease reported in companion animals (OMIA not definitive). Complex I deficiency broadly occurs across species.
- **Comparative biology / conservation:** The 14 core subunits are conserved from bacteria to humans; ~31 accessory subunits (incl. NDUFB7) are eukaryote-specific and integral for assembly/stability [Review;

  P 34069703

  P 27626371
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  Disease mechanism (energy failure → tissue degeneration) is evolutionarily conserved.
- **Transmission / zoonosis:** Not applicable (genetic, non-transmissible).

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15. Model Organisms

- **Cellular models:** Patient fibroblasts + wild-type *NDUFB7* complementation directly demonstrate the defect and rescue [In vitro;
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  P 33502047
  CRISPR knockout cell lines for each complex I accessory subunit define assembly requirements [In vitro;
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  P 27626371
  iPSC-derived cardiomyocytes/neurons and organoids are established for CI deficiency/Leigh broadly [Review;
  ].
  P 39385390
- **Mouse:** No published *Ndufb7* knockout disease model located; the canonical complex I / Leigh model is the **Ndufs4 knockout mouse**, which recapitulates progressive

encephalomyopathy, ataxia, and early death with region-specific respiratory deficits preceding symptoms [Model organism; P 34849584 P 18396137 P 26824698].

- **Invertebrate:** *Drosophila* complex I (ND2) mutants show shortened lifespan, neurodegeneration, low neural ATP, uncoupled proton pumping [Model organism; P 25085991]. *C. elegans*, zebrafish, and yeast are used across the CI/Leigh spectrum [Review; P 39385390].
- **Phenotype recapitulation & limitations:** Surrogate-gene models reproduce the neurodegenerative/energetic phenotype but not the NDUFB7-specific cardiac-predominant congenital presentation; an *Ndufb7*-specific model (and a cardiac model) is a key gap. Model limitations include species differences in lesion distribution/lifespan.
- **Resources:** MGI (mouse *Ndufb7*), ZFIN, FlyBase, Alliance of Genome Resources; IMPC/KOMP for knockout availability.

Supported vs. Refuted Hypotheses

- **Supported:** (1) MC1DN39 (OMIM 620135) is caused by biallelic loss-of-function *NDUFB7* variants — functional complementation confirms causality. (2) Mechanism = reduced NDUFB7 → destabilized complex I → isolated CI deficiency → ATP deficit/ROS → severe lactic acidosis + hypertrophic cardiomyopathy + encephalopathy. (3) Prognosis is poor (fatal in the reported case), matching severe nuclear CI deficiency.
- **Refuted / corrected:** The initial hypothesis that MC1DN39 = *NDUFB8* was **refuted**; NDUFB8 is MC1DN32 (OMIM 618252). Also refuted: environmental/infectious causation; gain-of-function/dominant mechanism; chromosomal-abnormality etiology; existence of a disease-modifying drug.

Limitations and Future Directions

- **Only one functionally proven family** (plus rare ClinVar variants) is reported for MC1DN39, so allele-frequency, penetrance, phenotype-frequency, and genotype–phenotype statements are severely limited; much is extrapolated from the broader complex I deficiency / Leigh literature.
- Phenotype description is dominated by a single congenital, cardiac-predominant, lethal case — the full clinical spectrum of NDUFB7 disease is unknown.
- No *Ndufb7*-specific animal model exists — a priority for mechanistic and preclinical (gene-replacement / splice-ASO) therapy work; the in-vitro rescue supports feasibility.
- Precise complex I structural position/module of NDUFB7 and its MIA40-dependent import are inferred from domain/DB annotation and were not confirmed by NDUFB7-specific structural literature in this report.

Key References (PMID)

- **33502047** — Correia et al. 2021. Severe congenital lactic acidosis and hypertrophic cardiomyopathy caused by an intronic NDUFB7 variant — **defining paper for MC1DN39**.
- 10830904 — Triepels et al. 2000. Characterization of human complex I NDUFB7 cDNA (historical gene context).
- 22644603 — Koene et al. 2012. Natural disease course of nuclear complex I deficiency (130 cases) — prognosis/phenotype.
- 27626371 — Stroud et al. 2016. Accessory subunits integral for assembly/function of human complex I.
- 34069703 — Kahlhöfer et al. 2021. Complex I structure/accessory subunits (review).
- 26363424 — de Haas et al. 2016. Leigh disease incidence 1/77,000; Ndufs4 model.
- 34849584 / 18396137 / 26824698 — Ndufs4 knockout mouse models of Leigh syndrome.
- 25085991 — Drosophila complex I (ND2) disease model.
- 20157008 — Neuronal mechanism: ROS, GSH depletion, reverse ATPase in complex I deficiency.
- 28943110 — El-Hattab et al. 2017. Therapies/clinical trials in mitochondrial disease.